

MA329 Statistical linear models 25-26

Assignment 2 (Due date: Oct 29, 11pm. For late submission, each day costs 10 percent)

1. (10 marks) If $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A} + \mathbf{PBQ}$ are nonsingular, prove that

$$(\mathbf{A} + \mathbf{PBQ})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{PB}(\mathbf{B} + \mathbf{BQA}^{-1}\mathbf{PB})^{-1}\mathbf{BQA}^{-1}$$

2. (15 marks) $\mathbf{A} = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}$

- (a) Find a symmetric generalized inverse for $\mathbf{A}$;
(b) Find a nonsymmetric generalized inverse for $\mathbf{A}$.
3. (15 marks) Prove that $(\mathbf{X}'\mathbf{X})^{-}\mathbf{X}'$ is a generalized inverse of $\mathbf{X}$ for any generalized inverse of $\mathbf{X}'\mathbf{X}$.
4. (20 marks) Let $\mathbf{X} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\mathbf{X}' = (X_1, X_2, X_3)$, $\boldsymbol{\mu}' = (3, -2, 0)$ and

$$\boldsymbol{\Sigma} = \begin{pmatrix} 5 & 0 & -3 \\ 0 & 9 & 0 \\ -3 & 0 & 2 \end{pmatrix}$$

- (a) Are X_2 and $2X_1 - X_3$ independent? Explain.
(b) Find the distribution of $\begin{pmatrix} 2X_1 - 5X_3 \\ X_1 + X_2 \end{pmatrix}$.
(c) Find the conditional distribution of X_3 , given that $X_1 = 1$ and $X_2 = -2$.
5. (10 marks) Let $\mathbf{Y} = (Y_1, Y_2, Y_3)'$. $E(\mathbf{Y}) = (2, 3, 4)'$ and the covariance matrix of $\mathbf{Y}$ is

$$\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & 3 \end{pmatrix}.$$

Let $U = \sum_{i=1}^3 (Y_i - \bar{Y})^2$. Find the expected value of U .

6. (10 marks) Let $\mathbf{Y} = (Y_1, \dots, Y_n)'$. $E(\mathbf{Y}) = \mu\mathbf{1}$ and the covariance matrix of $\mathbf{Y}$ is $\sigma^2\mathbf{I}$. Let

$$U = \sum_{i < j} (Y_i - Y_j)^2.$$

Find the expected value of U , and find a constant k such that kU is an unbiased estimator of σ^2 .